

Exploring Situated Instructions for Mixed Reality (MR) Remote Collaboration: Comparing Embedded and Non-Embedded Annotations

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Figure 1: Scenario of MR Remote Maintenance using two distinct conditions: C1 - Embedded and C2 - Non-Embedded. Devices used: Hand-Held Device (HHD-Left) and a Head-Mounted Display (HMD-Right).

ABSTRACT

Remote collaboration using Mixed Reality (MR) enable off-site experts to assist on-site collaborators needing guidance. Different types of visualizations have been proposed for sharing situated information, e.g., embedded, non-embedded, others. However, the effectiveness of these visualizations have not been compared. This work describes a user study with 16 participants, aimed at comparing two conditions: C1 - Embedded and C2 - Non-Embedded annotations during a real-life scenario of remote maintenance. Two distinct devices were used: Hand-Held Device (HHD) and a Head-Mounted Display (HMD). This last using Embedded annotations was considered the better alternative, while HHD using Non-Embedded annotations were rated lower from all conditions analyzed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Mixed/augmented reality;

1 INTRODUCTION

The research community has been exploring Mixed Reality (MR) to establish a common ground in scenarios of remote collaboration, with the goal of improving situation understanding, awareness and alertness [2, 3]. So far, most efforts have devoted to mature methods for Collaborative MR and to experiment with novel technological devices [1]. Although several approaches exist, literature suggests most studies have focused on the use of shared virtual annotations for augmenting the real-world environment. This is done by enhancing images, live video scenes, shared models/digital twins or physical reconstructions, either through drawings, pointers, gaze, gestures,

arrows or others [4]. In such contexts, it is possible to visualize shared annotations for the purpose of understanding, sense-making and decision-making in distinct manners: Embedded, i.e., having visual annotations deeply integrated with the context of the physical scenes to which the data refers to; Non-Embedded, i.e., having visual annotations displayed closer to the physical referents [6, 7]. However, exploring how these annotations are perceived by on-site collaborators and how they affect the overall collaborative process during such scenarios has not been properly explored.

2 USER STUDY

A user study was conducted to compare how different conditions: C1 - Embedded annotations; C2 - Non-Embedded annotations (Figure 1) affect on-site participants during a remote maintenance scenario supported by a MR-based tool. Two devices were considered: HHD - Hand-Held Device; HMD - Head-Mounted Display. The task was a defined-problem, i.e., instruct how to replace interconnected components of an equipment. To establish a common ground, an existing MR-based prototype was used [4]. Using audio communication and annotation features (e.g., drawing, placing pre-defined shapes or notes, sorting annotations), a remote expert could create MR annotations to guide the participants. These would receive enhanced pictures of the task context with specific areas of interest, questions or instructions, i.e., where to act, and what to do.

All data was collected through the *CAPTURE* toolkit [5], a package that can be added to collaborative solutions using Unity visual editor. It enables contextual data collection, providing additional perspective on selected aspects of collaboration, namely: D1- level of attentional allocation; D2- effectiveness in perceived information understanding; D3- level of mental effort; D4- level of satisfaction; participants preferences and opinions. Participants were instructed on the setup, the task, and gave their informed consent. Then, they were introduced to the MR-based tool. A within-group experimental design was used. Participants would act as on-site collaborators, while a researcher was the remote counterpart. Both conditions were used in different orders to minimize learning effects. At the end, the conditions were evaluated based on the dimensions considered and a small-interview occurred.

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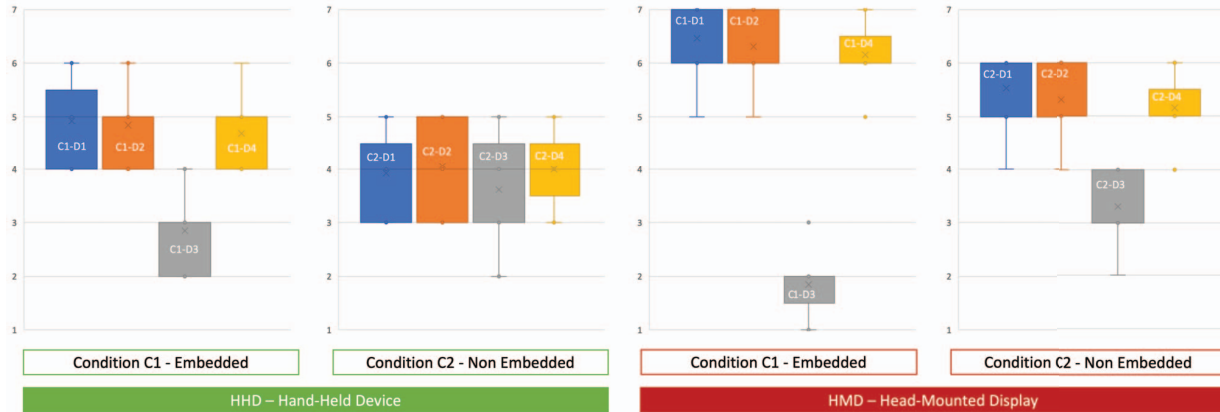


Figure 2: Overview of the collaborative process: D1- level of attentional allocation; D2- effectiveness in perceived information understanding; D3- level of mental effort; D4- level of satisfaction. Data displayed using a Likert-type scale: 1- Low; 7- High.

3 RESULTS AND DISCUSSION

Sixteen participants were recruited (5 female - 31.25%), whose ages ranged from 21 to 64 years old ($M=34.75$, $SD=12.70$). All participants had prior experience with MR tools and with common collaborative tools in general. Participants had various occupations, e.g., Master and PhD students, Researchers and Faculty members from different fields, as well as Software Engineers, a Manufacturing Manager, an Assembly Line Operator and a Housewife.

All in all, participants were able to fulfil the intended tasks with both conditions. When analysing the collaborative process in more detail, rated by the participants based on a Likert-type scale: 1- Low; 7- High (Figure 2), the following was reported. The **level of attentional allocation (D1)** was considered higher when using condition C1-HMD (median=7, sum=105), followed by condition C2-HMD (median=6, sum=88), condition C1-HHD (median=5, sum=77) and finally condition C2-HHD (median=4, sum=60). As to the **effectiveness in perceived information understanding (D2)**, similar results as the previous dimension were obtain: condition C1-HMD (median=6, sum=102), followed by condition C2-HMD (median=5, sum=84), condition C1-HHD (median=5, sum=79) and finally condition C2-HHD (median=4, sum=65). With respect to the **level of mental effort (D3)**, condition C1-HMD was rated lower (median=2, sum=29) followed by condition C1-HHD (median=3, sum=46), condition C2-HMD (median=3, sum=55) and finally condition C2-HHD (median=4, sum=62). Concerning the **level of satisfaction (D4)**, condition C1-HMD was rated higher (median=6, sum=100) followed by condition C2-HMD (median=5, sum=83), condition C1-HHD (median=5, sum=73) and finally condition C2-HHD (median=4, sum=65). During the interview, the majority of participants (14 out of 16) reported having preferred Embedded annotations and the HMD. It was emphasized that having the instructions aligned with the real-world environment, while also benefiting from a hands-free approach appears the most natural possibility to understand where to intervene and what needs to be done. Notwithstanding, 5 participants comment that having Embedded annotations in a HHD with an articulated support could improve their opinion towards this condition. Plus, 2 participants also mentioned the possible use of wrist-mounted displays as another relevant alternative.

4 CONCLUSIONS AND FUTURE WORK

Results from a user study with 16 participants suggest that both conditions may be considered as valid alternatives for guiding on-site individuals in need of assistance. Regardless, Embedded annotations (C1) appear the best alternative to visualize situated instructions,

in particular, when using a HMD, having obtain higher ratings for the dimensions: D1- level of attention allocation; D2- effectiveness in perceived information understanding; D4- level of satisfaction. Plus, it also demonstrated having lower levels of mental effort when compared to the remaining options.

As next steps, we intend to explore tasks with different levels of complexity to understand how participants may react to Embedded and Non-Embedded annotations for different types of use-cases. Also, explore how HHD with different screen sizes (distinct resolutions) may affect Non-Embedded annotations consumption and understanding. Furthermore, we aim to explore one-to-many scenarios, to understand if such type of annotations may play an important role when multiple remote experts are suggesting instructions at the same time, for the same on-site collaborator.

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